

SUGAM PANTHI

Undergraduate Researcher in Conversational Memory & LLM Evaluation

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RESEARCH INTERESTS

LLM evaluation: how measurement choices shape benchmark conclusions; scoring targets in conversational memory; and compression that distorts reader comparisons.

Tool-using agents: reliability under silent tool failure and recovery affordances for repairing corrupted results.

LLMs and human input: unmarked seams in a prompt, including the boundary between pasted material and user speech.

EDUCATION

The University of Southern Mississippi, Honors College Aug 2024 – May 2028 (Expected)

B.S. in Computer Science · GPA: 4.0/4.0

Hattiesburg, MS, USA

PUBLICATIONS

Peer-reviewed

[C1] **S. Panthi** and R. Abdelfattah, “Same Ranking, Different Winner: How Scoring Targets Shape LLM Memory Benchmarks,” *Findings of EMNLP*, 2026. arXiv:2605.24060

[C2] **S. Panthi** and F. Zhang, “A Comprehensive Review of Plastic Recycling in the Construction Industry: Challenges and Opportunities in the U.S.,” in *CIB Conferences*, vol. 1, art. 63, 2025. Purdue e-Pubs

Under review

[U1] **S. Panthi** and R. Abdelfattah, “Compression Is Not Evaluation-Neutral: Fixed RAG Compression Can Distort Reader Comparisons,” 2026. arXiv:2606.21807

[U2] **S. Panthi** and R. Abdelfattah, “Outcome Monitors: Recovery Affordances for Silent Tool Failures,” 2026. arXiv:2608.19303

[U3] **S. Panthi**, M. Yeamin, and R. Abdelfattah, “Can LLMs Separate Pasted Artifacts from User Speech? Absorption at Unmarked Prompt Seams,” 2026. arXiv:2610.04210

[U4] **S. Panthi**, M. Yeamin, S. Luo, and R. Abdelfattah, “Memory Consolidation Flattens the Temporal Shape of User Facts,” 2026. arXiv:2609.36457

RESEARCH EXPERIENCE

Scoring-Target Non-Invariance in LLM Memory Benchmarks [C1] Nov 2025 – Present

Lead Undergraduate Researcher; Advisor: Rabab Abdelfattah

- Found that LLM memory benchmarks can score the same retrieved item as a hit or miss depending only on which version of a fact they accept as correct. Named the problem target non-invariance: scores, system rankings, and design recommendations all move while the retrieved output stays fixed.
- Built an audit that rescores saved retrieval outputs under three definitions of correctness (Raw, Source, Canonical) without rerunning retrieval. On LoCoMo and LongMemEval-S, the definition alone changes nDCG on 83–94% of shared queries and reverses system rankings on Mem0 and MemoryOS. Tooling: Pali, the open-source memory runtime I maintain.

Outcome Monitors: Recovery Affordances for Silent Tool Failures [U2]

2026

Lead Undergraduate Researcher

- Built Outcome Monitors to catch tool calls that fail silently in LLM agents. Each result is checked against rules learned from unrelated successful runs; when a rule breaks, the agent is told which rule failed and which tools can recover.
- In frozen, prespecified evaluations with injected failures, raised ToolMaze task completion from 10.9% to 28.1% across four models from two providers, and τ -bench retail completion by 14.0 and 12.0 points on DeepSeek V4 Flash and V4 Pro. Ablations show the list of recovery tools is what does the work. Preprint: arXiv.

SIEVE: Post-Retrieval Evidence Compilation

2026

Sole Author

- Tested four ways of adapting how retrieved evidence is compressed before the model reads it, under one controlled pipeline with success and failure criteria fixed in advance. Only conditioning on the kind of query survived.
- Built a query-conditioned compiler that sits between BM25 retrieval and the reader model and keeps its gains across readers, avoiding the 30–40 point drops that one fixed compression policy causes. Builds on arXiv:2606.21807; code: GitHub.

SEAM: Absorption at Unmarked Prompt Seams [U3]

Jul 2026 – Present

Lead Undergraduate Researcher

- Built a controlled benchmark of matched editing tasks under different paste-boundary conditions. Across 20 models, a trailing user comment is absorbed into the pasted artifact 7.7–66.7% of the time at a bare new-line; a blank line does not help, explicit markers help in 19 of 20. Code: github.com/aimsresearchlab/seam.

LAPSE: Temporal Aspect Loss in LLM Memory Consolidation [U4]

2026

Lead Undergraduate Researcher; Advisor: Rabab Abdelfattah

- Built LAPSE, a benchmark of matched user statements that differ only in temporal form, to test whether memory writers keep the cue that a fact may not last (“I am driving a Peugeot” stored as “The user drives a Peugeot”). Three writers flattened the progressive while keeping its simple-present match in 244 of 381 pairs, never the reverse; the asymmetry holds in all 11 model configurations and in the installed mem0, Graphiti, and Letta pipelines. Preprint: arXiv.

Plastic-Waste Prediction with LSTMs [C2]

Nov 2024 – 2025

Undergraduate Research Assistant, The University of Southern Mississippi

- Modeled U.S. plastic-waste trends with LSTM networks; cleaned and visualized longitudinal datasets; authored methodology and results sections for two journal submissions.
- Built a React Native logistics app for recycling-plant waste transportation (Clerk, MongoDB, Mapbox).

INDUSTRY EXPERIENCE

Co-Founder & CTO, MagnoliaEd LLC

Apr 2025 – Present

- Built an AI-powered education platform with personalized instructor chatbots (RAG over course materials) and role-based dashboards; deployed across multiple college classrooms.
- Architected the system on AWS Fargate with Docker and Terraform, load-tested chat, analytics, and auth to 1,000+ concurrent sessions; CI/CD with GitHub Actions.
- Press: Innovate Mississippi, Mississippi Today.

AI & ML Fellow, Prediction 3D

May 2025 – Aug 2025

- Developed predictive models in PyTorch on multimodal construction-site data (images + sensor readings); integrated AI features into client applications with LangChain and Pinecone within a 20-engineer team.

Backend Intern, Best One Network

Feb 2024 – May 2024

- Rebuilt the payment-confirmation API (Django → Flask/WSGI); remediated 2 critical bugs pre-release, reducing post-deployment issues by 10%.

HONORS & AWARDS

Pre-Seed Funding, CoBuilders powered by Microsoft

2025

- Secured \$27,000 pre-seed funding for MagnoliaEd; led investor outreach.

3rd Place & People's Choice, MPI Pitch Competition

May 2025

Hackathon Winner, Davidson College

Jan 2025

Hackathon Winner (1st Place), USM Hatchathon

Nov 2024

SELECTED PROJECTS & OPEN SOURCE

Pali: Open-Source Memory Infrastructure for LLMs

GitHub

- Local-first, inspectable memory runtime for LLM agents in Go (REST, MCP, dashboard); multi-tenant lexical + vector retrieval with candidate fusion; pluggable backends (SQLite, Qdrant, Neo4j); benchmarked on LoCoMo, LongMemEval, and BEAM.

deepseek-go: Go SDK for DeepSeek-Compatible APIs

GitHub

- Lead maintainer (340+ GitHub stars, 35 forks), covering the full DeepSeek API surface; 45+ issues closed, 30+ PRs merged, releases through v1.3.3; used in production by multiple companies.

Cohesion: AI-Powered SQL Assistant (Hackathon Winner)

GitHub

- AI-driven schema generation with a speed-optimized CSV format (200% faster mock-data analysis); Go/Gin backend.

nomnom: GenAI Bulk File Renaming & Organization

GitHub

- CLI tool in Go for bulk renaming and organizing files with LLMs; distributed via a Homebrew tap.

LEADERSHIP & SERVICE

Head of Entrepreneurship Initiatives

2025 – Present

Google Developer Group on Campus, The University of Southern Mississippi

Hattiesburg, MS

- Led Semester Meetups on entrepreneurship, technical founding, and how undergraduate research begins.
- Co-organized three further GDG events, including a Robo Soccer tournament and chapter seminars.
- Mentored three first-generation undergraduates on early-stage software projects and research directions.

Executive Officer and Spokesperson

2025 – 2026

Nepalese Students Association, The University of Southern Mississippi

Hattiesburg, MS

- Organized cultural and community events drawing 300+ participants; handled logistics and volunteers.
- Served as spokesperson for the association, representing it to university administration.

TECHNICAL SKILLS

Languages: Python, Go, C/C++, TypeScript/JavaScript, SQL

ML/NLP: PyTorch, Hugging Face Transformers, LangChain, RAG pipelines, LLM/IR evaluation, fine-tuning, vector search (Pinecone, Qdrant)

Infrastructure & tools: AWS (ECS, Fargate, Lambda), Docker, Terraform, PostgreSQL, MongoDB, Neo4j, CI/CD, Linux, Git, L^AT_EX